Report on

Nvidia

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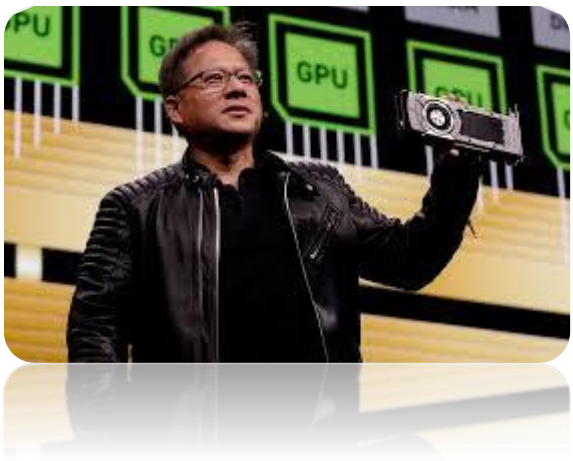
Introduction:

Nvidia Corporation is an American multinational technology company founded in 1993 and headquartered in Santa Clara, California. Renowned for its development of graphics processing units (GPUs), Nvidia has become a key player in the gaming industry with its GeForce GPU product line. The company also produces system-on-a-chip units for mobile devices and the automotive market.

In addition to its hardware products, Nvidia is at the forefront of artificial intelligence (AI) innovation. The company's AI technologies are widely used in various industries, including healthcare, automotive, and scientific research. With its strong focus on accelerated computing and AI, Nvidia continues to shape the future of technology.

Company overview :

Nvidia Corporation is an American multinational technology company headquartered in Santa Clara, California. Founded on April 5, 1993, by Jensen Huang**, Chris Malachowsky**, and Curtis Priem, Nvidia is best known for its graphics processing units (GPUs), which are widely used in gaming, professional visualization, data centers, and automotive markets.



Nvidia's GPUs are renowned for their performance and efficiency, making them a popular choice for high-end gaming PCs and professional workstations. The company's GeForce line of GPUs holds a dominant market share in the consumer market. Nvidia also produces system-on-a-chip (SoC) units for mobile devices and the automotive industry, including infotainment systems and autonomous driving solutions.

In addition to hardware, Nvidia is a leader in artificial intelligence (AI) technology. The company's AI platforms and software, such as CUDA and Omniverse, are used in various industries, including healthcare, automotive, and

scientific research. Nvidia's AI technologies are helping to drive advancements in fields such as medical imaging, autonomous vehicles, and climate modeling.

Nvidia continues to innovate and expand its product offerings, with a strong focus on accelerated computing and AI. The company's commitment to research and development has positioned it as a key player in the global technology industry.

Products of NVIDIA

Ideal for gamers and general users, the **GeForce series** includes:

- **GeForce RTX 50 Series**: Powered by the Blackwell architecture, featuring AI and neural rendering capabilities.
- **GeForce RTX 40 Series**: Known for ray tracing and AI technologies.
- **GeForce GTX 16 Series**: Popular for budget-friendly gaming options.

Quadro Series

Designed for professional graphic designers, video editors, and engineers:

- Quadro RTX Series: High-performance GPUs for professional workstations.
- Quadro M Series: Mid-range options for professional graphics.

Tesla and A100 Series

Focused on AI and research, these GPUs are used by data scientists and AI researchers:

- Tesla GPUs: High-performance computing solutions.

- A100 GPUs: Advanced AI and deep learning capabilities.

TITAN Series

Bridging the gap between GeForce and Quadro series, TITAN GPUs are designed for extreme performance:

- **TITAN RTX**: High-end gaming and professional graphics.

CUDA and Jetson

Expanding horizons in AI and embedded computing:

- **CUDA**: A parallel computing platform and programming model.

- **Jetson**: Embedded AI computing modules for robotics and IoT devices.

NVIDIA Drive

Steering the future of autonomous vehicles:

- **NVIDIA DRIVE AGX**: AI platforms for autonomous driving.

Omniverse

A platform for industrial digitalization and collaboration:

- **Omniverse Enterprise**: Enables real-time collaboration and simulation in various industries.

Revenue generation of NVIDIA:-

1. Gaming ($15-17 billion per year)

Percentage of Revenue: ~40-50%
Primary Products: GeForce GPUs, gaming laptops, gaming consoles
Revenue Model:

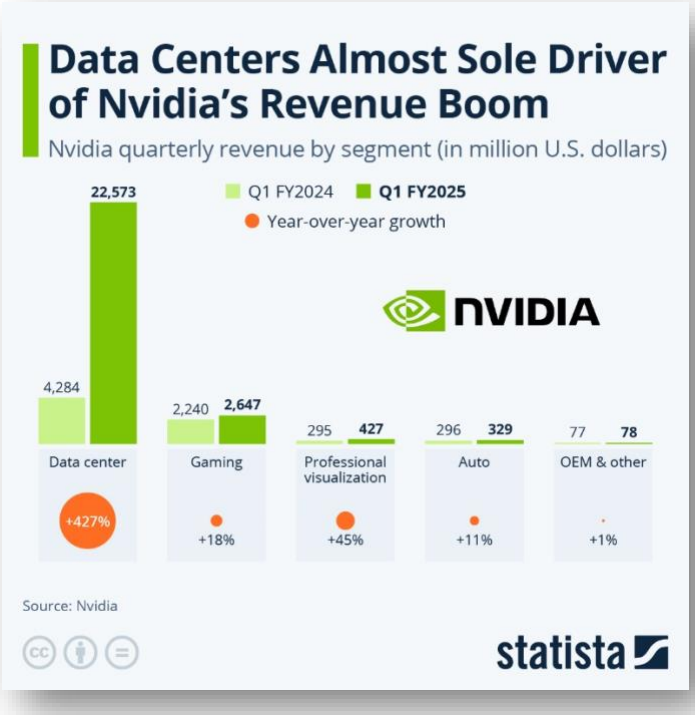
- **GeForce GPUs**: NVIDIA's primary revenue driver in gaming. Includes high-end cards (RTX 4090, 4080) and budget models (RTX 4060, GTX series).
- **GeForce NOW**: Cloud gaming subscription service (freemium and premium tiers).
- **NVIDIA Reflex & DLSS**: Software optimizations for gaming performance, included with GPUs.
- Gaming Laptops & Consoles: NVIDIA GPUs power gaming laptops and consoles like the Nintendo Switch.

2. Data Center & AI ($30-40 billion per year)

Percentage of Revenue: ~55-65%
Primary Products: AI GPUs, cloud computing solutions, networking
Revenue Model:

- **AI GPUs (H100, A100, L40, etc.)**: NVIDIA dominates AI acceleration, powering models like ChatGPT. Sold to tech giants (Google, Microsoft, Amazon, Meta, OpenAI).
- **NVIDIA DGX Systems**: AI supercomputing hardware for enterprises.
- **Cloud AI & Enterprise Licensing**: AI models and APIs (e.g., AI Foundry, CUDA, TensorRT).

- **Networking (Mellanox & Infiniband)**: High-speed data networking for AI workloads.
- **Hopper & Grace CPU+GPU architecture**: AI-specific chip designs for deep learning.



3. Automotive ($1-2 billion per year)

Percentage of Revenue: ~3-5%
Primary Products: Autonomous driving, infotainment chips, AI software
Revenue Model:

- **NVIDIA DRIVE**: AI-based self-driving platform used by Tesla, Mercedes, BYD, and others.
- **Infotainment Systems**: GPUs power dashboard displays, smart cockpits.

- **ADAS (Advanced Driver Assistance Systems)**: AI-driven safety features for vehicles.

4. Professional Visualization ($1-2 billion per year)

Percentage of Revenue: ~3-5%
Primary Products: Workstation GPUs for content creators, CAD, medical imaging
Revenue Model:

- **Quadro & RTX Workstation GPUs**: Used in video editing, 3D modeling (Autodesk, Adobe).
- **Omniverse**: AI-powered digital twin simulation & 3D collaboration software.
- **Medical & Scientific Visualization**: AI-powered imaging solutions in healthcare.

5. Licensing & Software ($1-2 billion per year)

Percentage of Revenue: ~3-5%
Primary Products: CUDA, AI models, cloud services, Omniverse
Revenue Model:

- **CUDA & AI Software Licensing**: Essential for AI and deep learning applications.
- **Omniverse Subscriptions**: Metaverse and industrial AI simulations.
- **Cloud-based AI APIs**: Paid APIs for AI model training and inference.

Financial report :

Fourth Quarter Fiscal 2024 (Q4 FY24)

- **Revenue**: ₹1,63,000 crore, up 22% from Q3 FY24 and up 265% from Q4 FY23.
- **GAAP Earnings Per Share (EPS)**: ₹3,500, up 33% from Q3 FY24 and up 765% from Q4 FY23.
- **Non-GAAP EPS**: ₹3,700, up 28% from Q3 FY24 and up 486% from Q4 FY23.
- **Gross Margin**: 76.0%, up 2.0 percentage points from Q3 FY24.
- **Operating Income**: ₹9,000 crore, up 31% from Q3 FY24.
- **Net Income**: ₹8,100 crore, up 33% from Q3 FY24.

Fiscal Year 2024 (FY24)

- **Total Revenue**: ₹5,00,000 crore, up 126% from FY23.
- **GAAP EPS**: ₹4,000, up 586% from FY23.
- **Non-GAAP EPS**: ₹4,300, up 288% from FY23.
- **Gross Margin**: 76.7%, up 1.7 percentage points from Q3 FY24.
- **Operating Income**: ₹9,500 crore, up 28% from Q3 FY24.
- **Net Income**: ₹8,300 crore, up 28% from Q3 FY24.

Investment in R&D

- **Annual R&D Expenses**: ₹59,000 crore, an 18.2% increase from FY23.
- **Quarterly R&D Expenses (Q4 FY24)**: ₹15,000 crore, a 47.78% increase year-over-year.

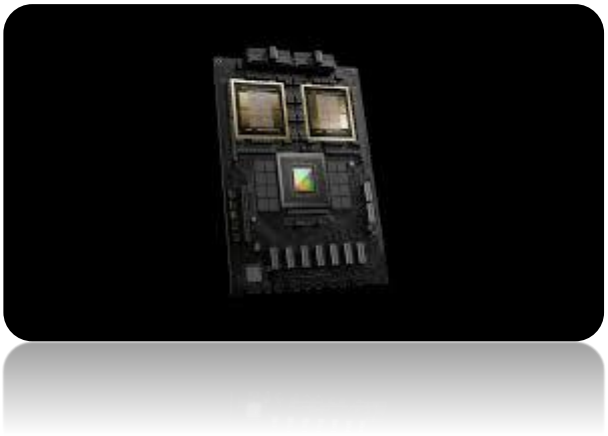
Stoke market report

As of February 10, 2025, Nvidia's stock price is approximately **₹9,500** per share. Nvidia's market capitalization is around **₹3,27,000 crore**.

NEW INNOVATION&TECNOLOGIES OF NVIDIA OF THE YEAR (2024-2025)

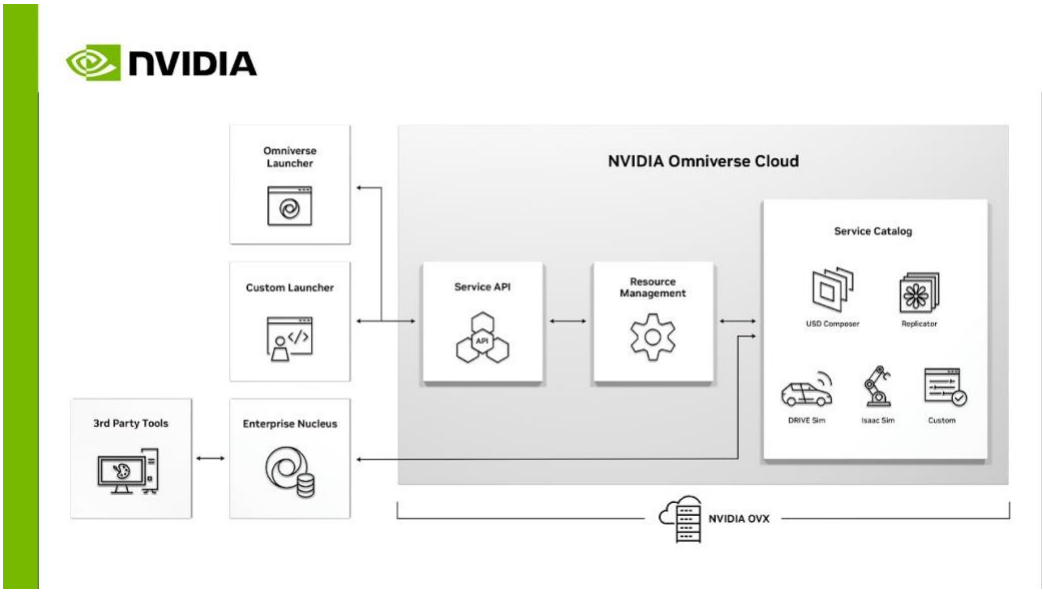
NVIDIA has introduced several innovations in recent years:

- **2024: Blackwell AI Platform**
  Unveiled at GTC 2024, the Blackwell AI platform features the B200 GPU, a dual-die architecture offering up to 30 times more inference throughput and four times better training performance than its predecessor, enhancing generative AI capabilities.



- **2024: Omniverse Cloud APIs**
  NVIDIA expanded its Omniverse platform by introducing Cloud APIs, enabling seamless integration of physical and virtual worlds for industries to simulate

and optimize operations efficiently.



- **2024: DRIVE Thor Platform**
  NVIDIA's DRIVE Thor platform, adopted by leading Chinese automakers, is designed for generative AI applications in vehicles, delivering up to a thousand trillion operations per second to enhance autonomous driving and in-car AI functionalities.

- **2025: Digits Personal AI Supercomputer**
  Set to launch in May 2025, Digits is a compact, $3,000 AI supercomputer equipped with NVIDIA's GB10 Grace Blackwell superchip, 128GB of unified memory, and up to 4TB of storage, enabling users to train and run large AI models without relying on data centers.

- **2025: RTX 50-Series GPUs**
  NVIDIA plans to release the RTX 5070 Ti and RTX 5070 , RTX 5090, RTX 5080, RTX 5070 Ti, and RTX 5070 GPUs in early 2025, offering affordable options for PC gamers to experience DLSS 4 features, including Multi Frame Generation for higher frame rates in compatible games.

- **2025: Studio Voice AI Tool**
  NVIDIA introduced 'Studio Voice,' an AI-based tool designed to enhance audio quality by processing sound in real-time, eliminating unwanted noise, and

optimizing clarity, effectively upgrading basic microphones to near-professional quality.

Competitors of NVIDIA:

1. **AMD (Advanced Micro Devices)**
   - Competes with NVIDIA in **GPUs (Radeon RX series)** and **Data Center AI chips (MI300X series)**.
2. **Intel**
   - Competes with **Arc GPUs for gaming** and **Gaudi AI chips** for data centers.
3. **Google (TPUs – Tensor Processing Units)**
   - Google's TPUs power its own AI models and compete with NVIDIA's AI GPUs (H100, B200)..
4. **Apple (Custom GPUs in M-Series chips)**

- Apple Silicon (M1, M2, M3) integrates **custom GPUs**, reducing dependence on NVIDIA for MacBooks.

5. **Microsoft & Amazon (Cloud AI Chips)**
    - Microsoft's **Azure Maia AI chip** and Amazon's **Trainium/Inferentia** challenge NVIDIA in cloud AI computing.
6. **Chinese AI Chipmakers (Huawei, Biren, Moore Threads, Tencent)**
    - **Huawei's Ascend AI chips** and **Biren BR100** aim to replace NVIDIA in China due to U.S. trade restrictions.

Future Outlook of NVIDIA (2025 & Beyond)

1. AI & Machine Learning

- Expansion of Blackwell AI GPUs (B200, H200) for deep learning & cloud computing.
- Growth in NVIDIA AI Enterprise software for businesses & cloud AI solutions.

- Increased adoption of Generative AI models & LLMs powered by NVIDIA TensorRT.

2. Gaming & GPU Innovation

- Launch of RTX 60-Series GPUs (Blackwell architecture) with DLSS 4 & AI-powered rendering.
- Expansion of GeForce NOW cloud gaming with AI-driven performance improvements.

- AI-powered real-time ray tracing, frame generation, & immersive gaming experiences.

3. Data Centers & Cloud AI

- AI-powered data centers driving NVIDIA's revenue growth in cloud computing.
- Collaboration with Google, Microsoft, Amazon for AI acceleration in cloud services.

- Competition from custom AI chips (TPUs, Gaudi,
- AWS Trainium) reducing reliance on NVIDIA.

4. Automotive & Robotics

- Expansion of NVIDIA DRIVE Thor AI for autonomous vehicles & AI-powered smart cars.

- Growth in AI robotics & industrial automation (Jetson AI Edge Computing).

5. Metaverse & Digital Twin Technology

- Increased adoption of NVIDIA Omniverse for 3D simulations, AI design, & enterprise solutions.
- Application in architecture, manufacturing, and healthcare digital twins.

7. Financial & Market Growth

- Expected annual revenue growth of 20-30%, driven by AI & data center demand.

- Strong stock market position, but facing volatility due to global semiconductor competition.

Conclusion

NVIDIA remains a leader in **AI, gaming, and data centers**, driven by innovation in **GPU and AI technology**. With upcoming advancements like **Blackwell AI GPUs (B200, H200), RTX 50-Series, and AI-powered cloud solutions**, the company is set for continued growth. Expanding into **enterprise AI, Omniverse, and autonomous vehicles** further strengthens its market position.

Financially, **record revenue growth** and an **expected 20-30% annual increase** highlight NVIDIA's strength. However, **U.S.-China trade restrictions, rising competition (AMD, Intel, cloud AI chips), and high GPU costs** pose challenges.

To maintain dominance, NVIDIA must **invest in AI, software, and energy-efficient GPUs**, while expanding into **robotics, automotive AI, and digital twins**. Despite competition, its **innovation, strong market position, and AI-driven strategy** secure its role as a key semiconductor leader.